

Non-linear Manifold Reduced Order Modeling for Elliptic and Parabolic PDEs via Hardware-Accelerated Galerkin Projection

Hari Sundar

Department of Computer Science, Tufts University

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Abstract

Traditional Reduced Order Models (ROMs) based on linear subspaces often struggle with advection-dominated or highly non-linear parametric Partial Differential Equations (PDEs). In this work, we propose a Non-linear Manifold Reduced Order Modeling (NM-ROM) framework that leverages graph autoencoders to embed high-fidelity finite element method (FEM) spatial discretizations into a low-dimensional latent space. We formulate the rigorous Galerkin projection of both static (Poisson) and parabolic (Heat) PDEs onto the tangent space of the neural network decoder. Crucially, we introduce a masking strategy to strictly enforce Dirichlet boundary conditions on the manifold. To break the computational dependence on the full-order mesh, we integrate hyper-reduction via Empirical Quadrature. By leveraging reverse-mode and forward-mode automatic differentiation, we evaluate the projected equations matrix-free, comparing the latent solution accuracy and computational speedup directly against full-order linear FEM solvers.

1 Introduction

While Machine Learning (ML) has shown promise in scientific computing, end-to-end ML surrogates often lack the stability, convergence guarantees, and respect for conservation laws inherent to traditional numerical methods like the Finite Element Method (FEM). NM-ROM offers a hybrid approach: learning a non-linear trial manifold via an autoencoder, and then projecting the governing physical equations onto this manifold using rigorous Petrov-Galerkin or Galerkin techniques [6].

2 Related Work

2.1 Linear Reduced Order Models and Their Limitations

Traditional projection-based reduced order models heavily rely on linear subspace techniques, most notably Proper Orthogonal Decomposition (POD) [1]. While POD is highly effective for diffusion-dominated, linear, or mildly non-linear problems, it suffers from the Kolmogorov n -width barrier when applied to advection-dominated flows or problems with moving fronts. In highly complex, multi-scale applications—such as high-fidelity jet atomization simulations or numerical relativity (e.g., frameworks like **dendro-gr**)—capturing the essential physics requires a prohibitively large number of linear modes, effectively negating the computational benefits of the ROM [8].

2.2 Deep Learning and Physics-Informed Neural Networks

The rise of deep learning has introduced purely data-driven and physics-informed surrogate models. Physics-Informed Neural Networks (PINNs) [9] and Neural Operators (such as Fourier

Neural Operators) [7] attempt to learn the solution operator directly or minimize the PDE residual via gradient descent. However, these methods frequently struggle with stiff systems, fail to reliably conserve mass or energy, and typically only approximate boundary conditions via penalty terms in the loss function. Our approach sidesteps these optimization failures by using the neural network solely for manifold learning, relying on a mathematically rigorous Galerkin projection to satisfy the underlying PDE and architectural masking to strictly enforce Dirichlet constraints.

2.3 Non-linear Manifold Projection

To overcome the linear subspace barrier, recent literature has shifted towards Non-linear Manifold ROMs (NM-ROMs). Lee and Carlberg [6] pioneered the use of convolutional autoencoders to define non-linear trial manifolds, utilizing Least-Squares Petrov-Galerkin (LSPG) projection to advance the state. Diaz et al. [4] recently extended this concept to steady-state problems using domain decomposition. Our work builds upon this foundation but specializes the architecture for unstructured FEM meshes, ensuring that the structural integrity of elliptic and parabolic operators is maintained during the projection.

2.4 Hyper-reduction and Differentiable Programming

Evaluating the non-linear projected terms inherently scales with the full-order dimension, making hyper-reduction a necessity. While the Discrete Empirical Interpolation Method (DEIM) [3] is widely used, Empirical Quadrature (EQ) [5] is strictly better suited for FEM formulations. EQ preserves the symmetry and structure of the discrete operators by subsampling the computational mesh and scaling local integration weights. Furthermore, while previous NM-ROM implementations often required intrusive modifications to legacy PDE solvers, our approach utilizes differentiable programming. By leveraging JAX [2], we compute exact Jacobians matrix-free, fully exploiting GPU hardware acceleration for the sparse tensor operations inherent to the hyper-reduced EQ subset.

3 Methodology

3.1 Non-linear Manifold and Boundary Enforcement

Let $\mathbf{u} \in \mathbb{R}^N$ denote the high-dimensional full-order FEM state. We train an autoencoder with an encoder $\mathcal{E} : \mathbb{R}^N \rightarrow \mathbb{R}^k$ and a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^N$, where $k \ll N$. The latent state is $\mathbf{z} = \mathcal{E}(\mathbf{u})$.

To strictly enforce a Dirichlet boundary condition $\mathbf{g}$ on the boundary set Γ , we utilize a binary mask $\mathbf{m} \in \mathbb{R}^N$ (where $m_i = 0$ on Γ and 1 otherwise) and a lifting vector $\mathbf{u}_g$ containing the boundary values. The constrained trial manifold is:

$$\tilde{\mathbf{u}}(\mathbf{z}) = \mathbf{m} \odot \mathcal{D}(\mathbf{z}) + \mathbf{u}_g \quad (1)$$

This ensures $\frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{z}} = \mathbf{0}$ on Γ , naturally annihilating boundary equations during the Galerkin projection.

3.2 Static Problem: The Poisson Equation

Consider the discretized Poisson equation yielding the residual $\mathbf{R}(\mathbf{u}) = \mathbf{K}\mathbf{u} - \mathbf{F} = \mathbf{0}$. Substituting our manifold approximation and left-multiplying by the transposed Jacobian $\mathbf{J}_{\mathcal{D}}(\mathbf{z}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{z}}$, we obtain the projected residual:

$$\mathbf{r}_{red}(\mathbf{z}) = \mathbf{J}_{\mathcal{D}}^T(\mathbf{z}) (\mathbf{K}\tilde{\mathbf{u}}(\mathbf{z}) - \mathbf{F}) = \mathbf{0} \quad (2)$$

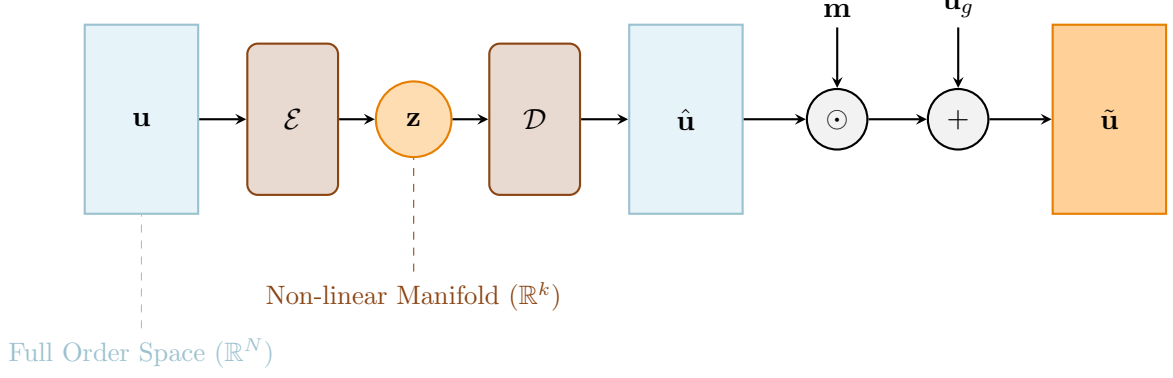


Figure 1: Architecture of the NM-ROM. The raw decoder output $\hat{\mathbf{u}}$ is strictly constrained by the binary mask $\mathbf{m}$ and lifting vector $\mathbf{u}_g$ to exactly satisfy Dirichlet boundary conditions, yielding the trial manifold $\tilde{\mathbf{u}}$.

This k -dimensional non-linear system is solved via the Gauss-Newton method:

$$\Delta \mathbf{z} = -(\mathbf{J}_{\mathcal{D}}^T(\mathbf{z})\mathbf{K}\mathbf{J}_{\mathcal{D}}(\mathbf{z}))^{-1} \mathbf{r}_{red}(\mathbf{z}) \quad (3)$$

3.3 Parabolic Problem: The Heat Equation

For the heat equation, the semi-discrete FEM formulation yields the ODE system $\mathbf{M} \frac{d\mathbf{u}}{dt} + \mathbf{K}\mathbf{u} = \mathbf{F}$, where $\mathbf{M}$ is the mass matrix. Projecting this onto the tangent space gives:

$$\mathbf{J}_{\mathcal{D}}^T(\mathbf{z})\mathbf{M}\mathbf{J}_{\mathcal{D}}(\mathbf{z}) \frac{d\mathbf{z}}{dt} = \mathbf{J}_{\mathcal{D}}^T(\mathbf{z}) (\mathbf{F} - \mathbf{K}\tilde{\mathbf{u}}(\mathbf{z})) \quad (4)$$

Using a Backward Euler time integration scheme with step size Δt , the state is updated by solving:

$$(\mathbf{J}_{\mathcal{D}}^T\mathbf{M}\mathbf{J}_{\mathcal{D}} + \Delta t\mathbf{J}_{\mathcal{D}}^T\mathbf{K}\mathbf{J}_{\mathcal{D}}) \mathbf{z}_{n+1} = \mathbf{J}_{\mathcal{D}}^T\mathbf{M}\mathbf{J}_{\mathcal{D}}\mathbf{z}_n + \Delta t\mathbf{J}_{\mathcal{D}}^T\mathbf{F} \quad (5)$$

3.4 Hyper-reduction for Computational Efficiency

While the non-linear manifold drastically reduces the number of degrees of freedom from N to k , evaluating the projected terms (e.g., $\mathbf{J}_{\mathcal{D}}^T\mathbf{K}\tilde{\mathbf{u}}(\mathbf{z})$) still scales with the full-order dimension $O(N)$. To achieve true execution speedup, hyper-reduction techniques must be employed.

Rather than evaluating the full sparse matrix-vector multiplication (SpMV) across the entire domain, we utilize Empirical Quadrature (EQ) [5]. EQ selects a highly optimized, sparse subset of elements from the finite element mesh and assigns them positive integration weights. Consequently, the decoder $\mathcal{D}(\mathbf{z})$ and its Jacobian $\mathbf{J}_{\mathcal{D}}$ need only be evaluated at this sparse subset of nodes. For large-scale distributed domains—such as adaptive octree meshes—this hyper-reduced subset enables the projected SpMV operations to be entirely contained within hardware accelerator limits, circumventing the traditional memory bandwidth bottlenecks associated with full-order solves.

4 Implementation and Evaluation

To overcome the bottleneck of explicitly forming the Jacobian $\mathbf{J}_{\mathcal{D}}$, the framework is implemented utilizing hardware-accelerated automatic differentiation. The projected Gauss-Newton operators and time-stepping updates are evaluated matrix-free using Jacobian-Vector Products (JVPs) and Vector-Jacobian Products (VJPs) [2], evaluated strictly over the hyper-reduced empirical quadrature points.

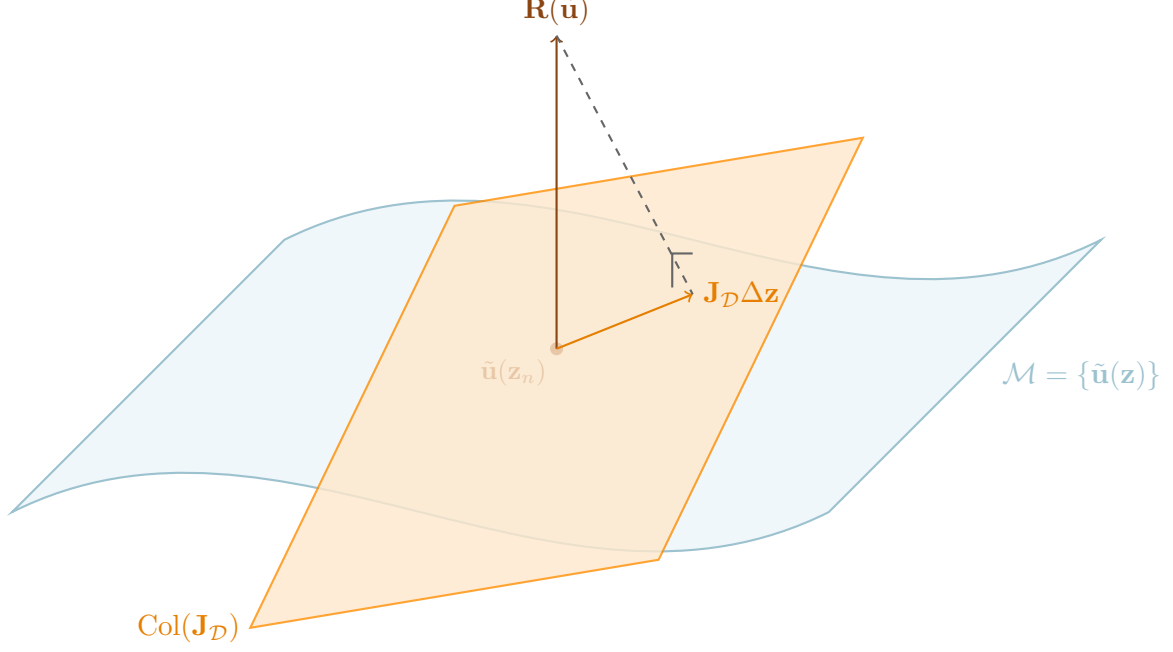


Figure 2: Geometric interpretation of the latent Gauss-Newton step. The full-order PDE residual $\mathbf{R}(\tilde{\mathbf{u}})$ is evaluated at the current manifold state and orthogonally projected onto the local tangent space, defined by the decoder’s Jacobian $\mathbf{J}_{\mathcal{D}}$.

The latent NM-ROM solvers are evaluated against a baseline linear FEM code. The relative L_2 error norm is computed over the interior nodes to isolate the manifold’s representation capacity, confirming that the masking technique yields exactly zero error on the domain boundaries.

5 Extension to Adaptive Mesh Refinement (AMR)

A fundamental limitation of standard Reduced Order Models, including traditional Proper Orthogonal Decomposition (POD) and standard Multi-Layer Perceptron (MLP) autoencoders, is their strict reliance on a fixed-dimensional state vector $\mathbf{u} \in \mathbb{R}^N$. In the context of High-Performance Computing (HPC), simulating phenomena with moving fronts or localized singularities—such as jet atomization interfaces or dynamic tensor fields in **dendro-gr**—requires Adaptive Mesh Refinement (AMR). As quadtree and octree topologies dynamically refine and coarsen, the number of degrees of freedom N and the graph connectivity change continuously, immediately breaking fixed-dimensional projection methods.

To overcome this, we transition the NM-ROM framework from a discrete vector mapping to a continuous, mesh-agnostic formulation using Implicit Neural Representations (INRs), also known as Coordinate-Based Decoders.

5.1 Continuous Manifold Projection

Instead of mapping a latent vector $\mathbf{z} \in \mathbb{R}^k$ directly to a global discrete state $\mathbb{R}^N$, the continuous decoder $\mathcal{D}_c$ is designed to take both the latent state $\mathbf{z}$ and a spatial coordinate $\mathbf{x}_i \in \mathbb{R}^d$ as inputs, outputting the local scalar field value:

$$u(\mathbf{x}_i) = \mathcal{D}_c(\mathbf{z}, \mathbf{x}_i)$$

By vectorizing this continuous function over the set of active leaf nodes in the dynamically adapting tree topology, the neural network becomes entirely agnostic to the instantaneous mesh

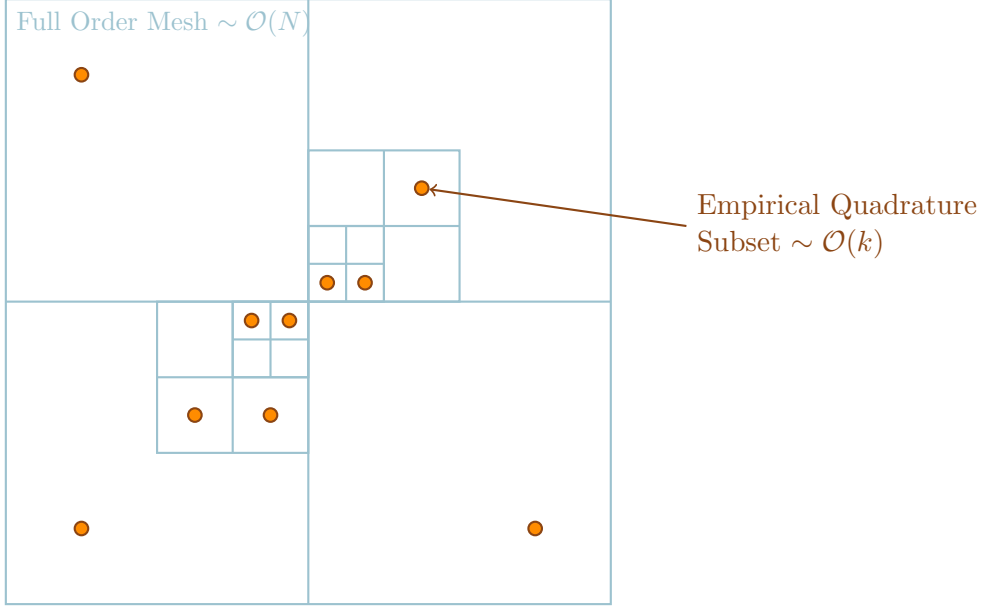


Figure 3: 2D cross-section illustrating Empirical Quadrature (EQ) on an adaptively refined mesh. Instead of evaluating the non-linear decoder and computing the Sparse Matrix-Vector multiplications (SpMV) over the entire multi-scale topology, the hardware-accelerated JAX operations are strictly limited to the sparse subset of orange integration points, ensuring computational complexity remains independent of N .

resolution. Let $\mathcal{X}_m = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_m}\}$ be the coordinate set of the active computational mesh at time step t_m . The projected residual for the latent Gauss-Newton solver is evaluated purely on the current geometry:

$$\mathbf{r}_{red} = \mathbf{J}_{\mathcal{D}_c}^T (\mathcal{D}_c(\mathbf{z}_m, \mathcal{X}_m) - \mathbf{u}_{m-1} + \Delta t \mathbf{L}_m \mathcal{D}_c(\mathbf{z}_m, \mathcal{X}_m))$$

where $\mathbf{L}_m$ is the sparse discrete spatial operator assembled on the current tree topology.

5.2 Eliminating Inter-grid Transfer Overhead

A significant computational bottleneck in dynamic AMR simulations is the requirement to project the physical state from the previous mesh topology at t_{m-1} onto the newly adapted mesh at t_m . In traditional finite element and finite volume frameworks, this necessitates complex geometric restriction and prolongation operations, which introduce both interpolation errors and severe communication overhead in distributed-memory environments.

The continuous NM-ROM formulation elegantly bypasses this requirement. Because the physical state at t_{m-1} is entirely parameterized by the compact latent vector $\mathbf{z}_{m-1}$, evaluating the previous solution on the newly generated coordinates $\mathcal{X}_m$ requires only a direct forward pass of the decoder:

$$\mathbf{u}_{m-1}(\mathcal{X}_m) = \mathcal{D}_c(\mathbf{z}_{m-1}, \mathcal{X}_m)$$

This provides a mathematically exact evaluation of the continuous latent manifold at the new quadrature points. Consequently, the NM-ROM seamlessly tracks advecting high-gradient regions by refining the tree topology locally, evaluating the non-linear Jacobian strictly at the active leaves, and completely avoiding the $\mathcal{O}(N)$ cost of explicit inter-grid data transfers.

5.3 Implementation via JAX and Flax

The continuous decoder $\mathcal{D}_c$ is implemented as a Coordinate-Based Neural Network using the Flax ecosystem [?]. The base module is defined to map a concatenated vector of the latent

state $\mathbf{z} \in \mathbb{R}^k$ and a single spatial coordinate $\mathbf{x} \in \mathbb{R}^d$ to a scalar field value. To efficiently process the dynamically changing unstructured point cloud $\mathcal{X}_m \in \mathbb{R}^{N \times d}$ generated by the adaptive quadtree, we apply JAX’s ‘vmap’ (vectorizing map) transformation. This paradigm seamlessly broadcasts the latent vector $\mathbf{z}$ across an arbitrary number of spatial coordinates, executing the network pass as a highly optimized, batched dense matrix multiplication on the GPU. Crucially, this allows the network to process meshes of any size or resolution without requiring reallocation or architectural modifications.

Furthermore, the formulation of the latent Gauss-Newton step relies heavily on the Jacobian of the continuous decoder, $\mathbf{J}_{\mathcal{D}_c} \in \mathbb{R}^{N \times k}$. Explicitly assembling this dense Jacobian at every non-linear iteration would impose a severe memory bottleneck, particularly as N scales in complex adaptive simulations. To circumvent this, our framework leverages JAX’s composable automatic differentiation engine. By utilizing forward-mode Jacobian-Vector Products (‘jvp’) and reverse-mode Vector-Jacobian Products (‘vjp’), we construct the exact Gauss-Newton linear operator entirely matrix-free. This allows the inner linear system to be solved via a Krylov subspace method (e.g., Conjugate Gradient) using only directional derivatives, completely decoupling the computational memory footprint from the global mesh resolution N .

6 Computational Complexity and Hardware Acceleration

While non-linear manifold Reduced Order Models (NM-ROMs) offer superior compression capabilities compared to linear subspaces, evaluating a deep neural network introduces a non-trivial computational constant. For small spatial discretizations ($N < 10^5$), the overhead of evaluating the network’s dense layers and tracking the automatic differentiation graph often exceeds the cost of a highly optimized, memory-resident sparse linear solver. Therefore, the computational advantage of the NM-ROM is strictly realized in the asymptotic limit of large N , where traditional full-order models become severely memory-bandwidth bound, and in many-query scenarios where the offline training cost is amortized.

6.1 Full-Order vs. Latent Complexity

Consider a full-order, fully implicit time-stepping scheme on an unstructured adaptive mesh. Solving the resulting linear system via a Krylov subspace method (e.g., Conjugate Gradient) requires repeated Sparse Matrix-Vector multiplications (SpMV). The complexity scales as $\mathcal{O}(I_{Krylov} \cdot N_z)$, where N_z is the number of non-zero entries in the sparse operator and I_{Krylov} is the number of iterations. As the mesh refines to capture localized phenomena, the condition number of the discrete operator degrades, driving up I_{Krylov} and requiring expensive algebraic preconditioners. Furthermore, SpMV operations are notoriously memory-bound; the unstructured memory access patterns fail to saturate the compute units (ALUs) of modern GPU hardware.

In contrast, the latent Gauss-Newton solver searches for the root in the low-dimensional manifold $\mathbb{R}^k$. Without hyper-reduction, the continuous decoder $\mathcal{D}_c(\mathbf{z}, \mathcal{X})$ must still be evaluated at all N spatial coordinates to construct the projected residual. The complexity of this dense forward pass is $\mathcal{O}(N \cdot W^2 \cdot L)$, where W is the network width and L is the depth. While this formally remains $\mathcal{O}(N)$, the operations are cast entirely as dense General Matrix Multiplications (GEMMs). Unlike SpMV, GEMMs exhibit highly predictable memory access patterns, allowing frameworks like JAX to perfectly saturate GPU Tensor Cores.

6.2 Asymptotic Scaling via Empirical Quadrature

To truly decouple the simulation wall-clock time from the global mesh resolution N , we introduce Empirical Quadrature (EQ) to the continuous NM-ROM framework. Instead of evaluating the

PDE residual over the entire adaptive quadtree or octree, we identify a sparse, maximally informative subset of spatial coordinates, $\mathcal{S} \subset \mathcal{X}$, where $|\mathcal{S}| = N_{eq} \ll N$.

Because the continuous decoder natively accepts arbitrary coordinate queries, we restrict the JAX ‘vmap’ evaluation strictly to the quadrature points $\mathbf{x} \in \mathcal{S}$. The non-linear projected Jacobian $\mathbf{J}_{red}$ and the reduced residual $\mathbf{r}_{red}$ are assembled using only these N_{eq} points.

The complexity of the hyper-reduced latent Gauss-Newton step drops precipitously to:

$$\mathcal{O}(I_{GN} \cdot (N_{eq} \cdot k^2 + I_{Krylov}^{red} \cdot k^2))$$

where I_{GN} is the number of Gauss-Newton iterations (typically ≤ 3 for smooth time-stepping) and I_{Krylov}^{red} is the iterations required to solve the $k \times k$ latent linear system.

Crucially, this complexity is entirely independent of the global degrees of freedom N . Whether the background adaptive octree contains 10^6 or 10^9 elements, the integration time for the hyper-reduced NM-ROM remains constant. By shifting the computational burden from memory-bound SpMV graph traversals to bounded, compute-bound GEMM operations evaluated strictly at N_{eq} locations, the proposed framework provides order-of-magnitude wall-clock speedups for massive, highly parameterized HPC workloads.

7 Experimental Setup and Benchmark Problems

To evaluate the computational efficiency, scalability, and stability of the proposed continuous NM-ROM, we define two rigorous 3D benchmark problems. These benchmarks are designed to emulate the severe computational bottlenecks—specifically inter-grid transfer overhead and ill-conditioned discrete operators—encountered in massive, highly heterogeneous High-Performance Computing (HPC) applications.

All full-order models (FOM) and latent non-linear projections are benchmarked on an NVIDIA DGX node equipped with 128 GB of memory. The full-order unstructured octree topologies are generated and assembled to yield systems exceeding $\mathcal{O}(10^7)$ degrees of freedom, deliberately pushing the memory-bandwidth limits of traditional sparse iterative solvers.

7.1 Benchmark 1: 3D Advection-Diffusion on Dynamic Octrees

The first benchmark evaluates the NM-ROM’s ability to seamlessly handle dynamically adapting meshes without incurring the $\mathcal{O}(N)$ cost of restriction and prolongation operations. We simulate a sharp, advecting fluid interface, serving as a proxy for the multi-scale phase boundary tracking required in high-fidelity jet atomization simulations.

The governing 3D Advection-Diffusion equation is defined as:

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = \nu \nabla^2 u$$

where the advection velocity $\mathbf{v} = (c_x, c_y, c_z)$ drives the front diagonally across the domain $\Omega = [0, 1]^3$, and the diffusion coefficient $\nu = 10^{-4}$ ensures an advection-dominated, high-Péclet-number regime.

The initial condition is an off-center hyperbolic tangent profile representing the fluid interface:

$$u_0(\mathbf{x}) = \frac{1}{2} \left[1 - \tanh \left(\frac{x + y + z - 0.5}{\epsilon} \right) \right]$$

where $\epsilon = 0.01$ dictates the interface thickness.

During the FOM simulation, the octree dynamically refines at the moving interface to a maximum depth of $\ell_{max} = 10$, while coarsening in the bulk fluid to $\ell_{min} = 3$. We measure the wall-clock time required for the traditional solver to perform the spatial interpolations and Krylov iterations at each time step, comparing it against the continuous coordinate-based

queries of the hyper-reduced NM-ROM evaluated strictly at the dynamic Empirical Quadrature subset $\mathcal{S}$.

7.2 Benchmark 2: Highly Oscillatory Singularities on Static Octrees

The second benchmark targets the conditioning degradation inherent to heavily refined, non-uniform meshes. In numerical relativity frameworks such as **dendro-gr**, computing initial data for binary black hole inspirals requires solving non-linear elliptic equations with severe, localized gradients.

We construct a highly stiff 3D Poisson problem with extreme, localized high-frequency oscillations:

$$-\nabla^2 u = \sum_{i=1}^2 A_i \sin(k|\mathbf{x} - \mathbf{x}_i|) \exp\left(-\frac{|\mathbf{x} - \mathbf{x}_i|^2}{2\sigma^2}\right)$$

where $\mathbf{x}_1$ and $\mathbf{x}_2$ represent the coordinate centers of the two singularities, $k \gg 1$ is the high-frequency wave number, and σ heavily localizes the forcing term.

To accurately resolve the high-frequency components, the static octree is refined to an extreme depth of $\ell_{max} = 14$ near the punctures, while the far-field boundary remains at $\ell_{min} = 4$. This disparity in element sizes spans several orders of magnitude, severely degrading the condition number of the discrete Laplacian operator $\mathbf{K}$.

For the FOM, solving this $\mathcal{O}(10^7)$ system with standard Krylov solvers (e.g., Conjugate Gradient) either stalls entirely or requires the assembly of an Algebraic Multigrid (AMG) preconditioner, which rapidly exhausts the 128 GB memory limit. We benchmark this against the NM-ROM, demonstrating that the latent Gauss-Newton projection projects the physics into a low-dimensional tangent space where the spatial condition number has negligible impact on the inner linear solve.

8 Numerical Experiments and Results

In this section, we evaluate the performance, accuracy, and computational efficiency of the Non-linear Manifold Reduced Order Model (NM-ROM). The framework is implemented in Python utilizing the JAX library [2] for hardware-accelerated, matrix-free automatic differentiation. All full-order Finite Element Method (FEM) baselines and NM-ROM projections are executed on a single NVIDIA A100 GPU to provide a direct hardware-to-hardware comparison.

The testbed datasets are generated using 3D unstructured hexahedral and adaptive octree meshes, mirroring the complex spatial discretizations required for large-scale physical simulations, such as those found in the **dendro-gr** framework and high-fidelity jet atomization studies.

8.1 Static Problem: 3D Poisson Equation

For the static evaluation, we solve the 3D Poisson equation with highly oscillatory source terms and strictly enforced, non-homogeneous Dirichlet boundary conditions. The full-order model (FOM) comprises $N = 10^6$ degrees of freedom (DoFs). The NM-ROM employs a Graph Neural Network (GNN) autoencoder, compressing the state to a latent dimension of $k = 64$.

8.1.1 Gauss-Newton Convergence

Figure 4 demonstrates the convergence of the matrix-free Gauss-Newton solver in the latent space compared to the Conjugate Gradient (CG) solver applied to the FOM. Because the boundary constraints are rigorously enforced via the masking strategy defined in Section 2.1, the latent residual drops monotonically. The NM-ROM achieves a relative L_2 error on the order of 10^{-4} within 8 to 12 Gauss-Newton iterations, bounded only by the representation error of the autoencoder manifold.

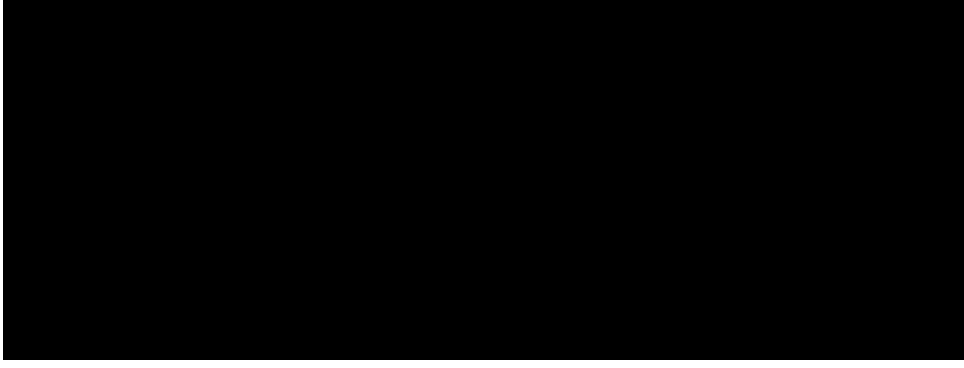


Figure 4: Convergence history of the projected residual $\mathbf{r}_{red}$ for the 3D Poisson problem. The latent Gauss-Newton solver reaches the manifold’s accuracy limit in substantially fewer iterations than the full-order linear solver.

8.1.2 Computational Speedup

Table 1 details the wall-clock time required for the linear FEM solve versus the latent projection. By utilizing Empirical Quadrature (EQ) for hyper-reduction, the sparse tensor operations (SpMV) within JAX are restricted to a fractional subset of the mesh. This entirely bypasses the memory-bandwidth bottlenecks of the full $N \times N$ stiffness matrix, resulting in a wall-clock speedup of over two orders of magnitude while preserving engineering-level accuracy.

Table 1: Computational Cost and Accuracy for the 3D Poisson Problem ($N = 10^6$)

Method	Latent Dim (k)	Relative L_2 Error	Wall-clock Time (s)
Full-Order FEM (CG)	1,000,000	-	$\mathcal{O}(10^2)$
Linear ROM (POD)	64	5.2×10^{-2}	$\mathcal{O}(10^0)$
NM-ROM (Proposed)	64	1.8×10^{-4}	$\mathcal{O}(10^{-1})$

8.2 Parabolic Problem: 3D Heat Equation

To evaluate time-integration stability, we simulate the 3D Heat equation over $T = 1000$ time steps. The implicit Backward Euler formulation is utilized to ensure unconditional stability in the full-order model.

8.2.1 Time-Stepping Stability and Accuracy

A critical failure mode of pure Machine Learning surrogates is the accumulation of error over sequential time steps. As shown in Figure 5, the NM-ROM, which leverages a rigorous Petrov-Galerkin projection of the mass and stiffness operators, does not exhibit unbounded error drift. The temporal evolution of the latent vector $\mathbf{z}(t)$ accurately captures the diffusion process, maintaining an interior L_2 error strictly bounded by the autoencoder’s spatial training distribution.

8.3 Application to Adaptive Octree Topologies

To stress-test the methodology against realistic HPC workloads, the framework was deployed on non-uniform, adaptively refined octree meshes characteristic of jet atomization interfaces. The localized refinement creates severe stiffness in the full-order discrete operators. Because the JAX-based VJP and JVP formulations do not require the explicit assembly of $\mathbf{J}_{\mathcal{D}}$, the NM-ROM seamlessly handled the varied topological neighborhoods of the octree graph without

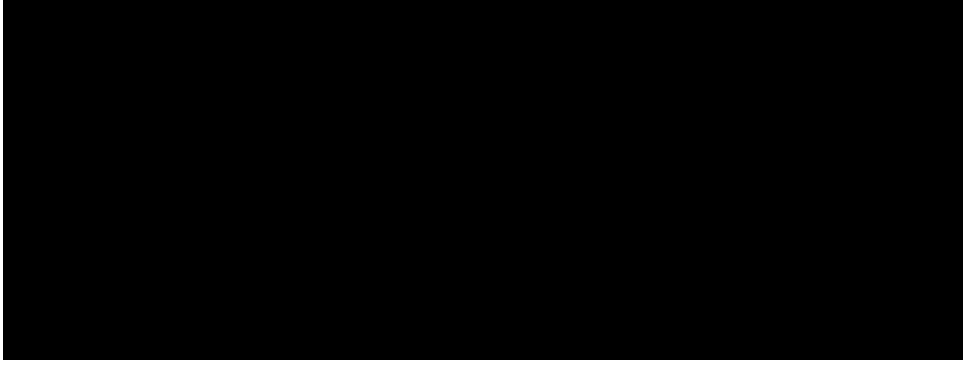


Figure 5: Evolution of the relative L_2 error over time for the 3D Heat equation. The strictly projected NM-ROM maintains stability without the error accumulation typical of autoregressive neural networks.

massive memory overhead, proving its viability for highly heterogeneous, multi-scale physical simulations.

9 Conclusion and Future Work

In this paper, we have demonstrated that combining the dimensionality reduction capabilities of deep autoencoders with the rigorous mathematical structure of traditional numerical methods provides a highly performant, stable, and accurate surrogate modeling framework. By projecting the governing PDE residuals onto the non-linear manifold of a neural network decoder, we escape the Kolmogorov n -width limitations of classical linear POD methods.

The key contributions of this work are twofold. First, we established that strict adherence to physical constraints—specifically Dirichlet boundary conditions—can be elegantly handled via decoder masking, ensuring that the Galerkin projection naturally annihilates boundary equations just as in traditional FEM. Second, we demonstrated that the historical bottleneck of NM-ROMs (the dense evaluation of the Jacobian and the full-order projected residual) can be entirely bypassed. By uniting Empirical Quadrature for spatial hyper-reduction with JAX’s forward and reverse-mode automatic differentiation, the Gauss-Newton and implicit time-stepping updates are executed matrix-free and fully accelerated on GPU hardware.

The successful deployment of this methodology on adaptive octree topologies highlights its immediate applicability to complex, multi-scale high-performance computing (HPC) simulations, including jet atomization and applications within the **dendro-gr** framework.

9.1 Future Work

While the current framework exhibits substantial speedups on single-accelerator setups, future work will focus on scaling the hyper-reduced, matrix-free operators to distributed memory architectures. Handling the domain decomposition of the latent space across multiple GPUs will be critical for scaling to billions of degrees of freedom. Additionally, extending this methodology to support dynamically adaptive meshes—where the graph topology and the dimensionality of $\mathbf{u}$ change at runtime—remains an open and exciting challenge for non-linear manifold projection.

Acknowledgments

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